CANON : Pre-Publication Brand Governance

CodeNexus India 2026 | Team RUNTIME REBELS | Technical Sprint Manifest

Target Hackathon: CodeNexus India 2026	**Sprint Date:** Tomorrow Morning (Pre-Pitch)
Problem: Inkloom AI Brand Governance	**Status:** Core Foundation 100% Tested & Verified

Sprint Objective & Architecture Contract

This execution document assigns **3 Core Required Functions + 2 Optional Expansion Functions** to each of the 3 teammates (Teammate 1, Teammate 2, and Teammate 3). Every feature has been surgically isolated to specific components and APIs to guarantee **zero code collisions and zero merge conflicts**. Each function is defined with its user persona, interaction step-by-step, enterprise business impact, and exact pitch talking points for the judges.

TEAMMATE 1 — Frontend & Interactive Authoring Specialist

Domain Ownership: UI/UX polish, copywriter experience, client feedback, print artifacts, and navigation speed.

Function 1.1: Official Printable PDF/HTML Compliance Release Certificate

[CORE REQUIRED]

Target Persona: Brand Compliance Officer / Enterprise Legal Team

Source Files: `frontend/src/components/ComplianceCertificateModal.tsx`

Function Summary: Add an official print stylesheet / PDF renderer to ComplianceCertificateModal.tsx that outputs an elegant, formal compliance certificate layout with the official seal, SHA-256 hashes, approver stamp, and timestamp.

How It Works: User clicks '[Compliance Certificate]' on any approved draft in /review or clean copy in /submissions, opens the modal, and clicks '[Print / PDF]'. Browser triggers clean native print window formatted to A4/Letter size with high-contrast borders and zero website nav junk.

Enterprise Benefit: Gives corporate clients an immutable physical or digital artifact that can be filed in regulatory compliance dossiers before an ad or article is pushed live.

Judge Pitch Impact: **During pitch: 'Judges, Canon does not just check text; it generates a formal cryptographic Release Certificate ready for audit trails.'**

Function 1.2: Real-Time Word/Char Counter & Channel Scope Limits

[CORE REQUIRED]

Target Persona: Content Creator / Copywriter

Source Files: `frontend/src/app/submissions/page.tsx`

Function Summary: Integrate live character count, word count, estimated reading time, and channel-specific length constraints into the authoring view (/submissions).

How It Works: While typing or uploading drafts in /submissions, dynamic badges display '248 words | 1,420 chars | 1.2 min read'. If channel is 'ad_copy' and chars > 280, a high-visibility badge alerts: 'Exceeds standard 280-char ad length constraint'.

Enterprise Benefit: Prevents creators from writing copy that violates channel distribution guidelines (e.g., character truncation on Twitter/LinkedIn, email preview cutoffs).

Judge Pitch Impact: **Demonstrates deep UX attention to real marketing workflows rather than a generic text box.**

Function 1.3: Global Quick Command Palette (Cmd+K / Ctrl+K)

[CORE REQUIRED]

Target Persona: All Users & Hackathon Judges during Live Demo

Source Files: `frontend/src/components/CommandPalette.tsx, frontend/src/app/Navbar.tsx`

Function Summary: A keyboard-accessible neo-brutalist modal dialog opened via Cmd+K or Ctrl+K that provides instant search and navigation across all views, test fixtures, and personas.

How It Works: Pressing Cmd+K anywhere brings up a sleek dialog. Typing 'SUB-02' loads the 4-violation sample draft; typing 'eval' jumps to /evaluation; typing 'reviewer' switches persona to Rachel Reviewer.

Enterprise Benefit: Eliminates menu clicking during presentations; provides a blazing fast Linear-style power user interface.

Judge Pitch Impact: **Judges love seeing power-user shortcuts: 'We built a global command palette so brand managers can triage thousands of assets without taking hands off the keyboard.'**

Function 1.4: Sent-Back Revision Diff Visualizer

[OPTIONAL BONUS]

Target Persona: Brand Reviewer & Copywriter

Source Files: `frontend/src/app/review/page.tsx`

Function Summary: In the /review sign-off tab, when inspecting a submission that was previously sent back, render an inline red/green word diff comparing Version 1 to Version 2.

How It Works: Reviewer sees exactly what the creator edited in response to the send-back notes, highlighting added words in green and removed violations in red strikethrough.

Enterprise Benefit: Cuts review triage time by 80% because reviewers don't have to re-read the entire document.

Judge Pitch Impact: **Visually demonstrates how Canon closes the loop between review feedback and author remediation.**

Function 1.5: Interactive Scan Audio / Micro-Haptic Feedback

[OPTIONAL BONUS]

Source Files: `frontend/src/lib/audio.ts, frontend/src/app/submissions/page.tsx`

Target Persona: Author / Reviewer

Function Summary: Add subtle audio cues (soft mechanical click on check, clean chime on pass, gentle buzz on critical violation) toggled via a sound icon in the navbar.

How It Works: User clicks 'Run Brand Check'. Upon receipt of response, Web Audio API produces a subtle sound matching the outcome. Sound is muted by default with an audio toggle.

Enterprise Benefit: Increases visceral satisfaction and polish during demo interactions.

Judge Pitch Impact: **Shows top-tier micro-interaction craft expected in modern products.**

TEAMMATE 2 — Domain, Policy & Legal Compliance Specialist

Domain Ownership: Regulatory defensibility, claims substantiation ledger, statutory legal tooltips, and cryptographic audit proofs.

Function 2.1: Interactive Claims Ledger Expiry Simulator [CORE REQUIRED]

Source Files: `frontend/src/app/policy/page.tsx, backend/app/api/v1/governance.py`

Target Persona: Legal Counsel / Brand Governance Lead

Function Summary: A dedicated Claims Ledger viewer on /policy that displays all claims ('10x Faster', '#1 in Enterprise') with substantiation windows, plus a live 'Fast-Forward Time' toggle (+6 Months).

How It Works: User opens Claims Ledger, sees valid claims. Clicks '+6 Months Time Travel'. The claims ledger updates expiry status to EXPIRED. Returning to /submissions and running SUB-01 now triggers a Critical Expired Claim violation!

Enterprise Benefit: Demonstrates the Temporal Claims Ledger Invariant: claims that are legal today become illegal tomorrow if substantiation studies lapse.

Judge Pitch Impact: **This is a massive killer feature against competitor spell-checkers: 'Generic AI does not understand temporal claim validity; Canon mathematically enforces time-windowed legal substantiations.'**

Function 2.2: Statutory Regulatory Badges (FTC, EU AI Act, SEC) [CORE REQUIRED]

Source Files: `data/rules.json, frontend/src/app/policy/page.tsx`

Target Persona: Brand Compliance Officer & Hackathon Judges

Function Summary: Attach real statutory citations and legal liability badges to rule cards in /policy and violation pills in /submissions (e.g., FTC Green Guides § 260.4, EU AI Act Art. 14, SEC Rule 206(4)-1).

How It Works: Hovering or clicking on a rule badge opens an explainer card showing the exact legal statute, regulatory jurisdiction, and statutory fine exposure (e.g., '$51,744 per day under FTC Act Section 5').

Enterprise Benefit: Transforms Canon from an arbitrary 'grammar linter' into a high-stakes enterprise regulatory shield.

Judge Pitch Impact: **Directly answers the enterprise buyer question: 'Why does our legal department need this platform?'**

Function 2.3: Cryptographic Audit Trail Hash Chain Verifier [CORE REQUIRED]

Source Files: `frontend/src/app/transparency/page.tsx, backend/app/api/v1/governance.py`

Target Persona: Internal & External Enterprise Auditors

Function Summary: On /transparency, add a 'Verify Ledger Integrity' button that traverses all audit logs, calculates the SHA-256 hash of each row against the previous row's hash, and visually displays cryptographic proof of non-tampering.

How It Works: Auditor clicks 'Run Cryptographic Audit'. An animated verification progress checks block hashes in 10ms and displays: '34 / 34 Blocks Verified — Zero Tampering Detected — Root Hash: 8b1f...'.

Enterprise Benefit: Proves Canon's 3rd Core Invariant: Immutable append-only audit trail with SHA-256 hash chaining.

Judge Pitch Impact: **Demonstrates institutional trust: 'Even if an admin has direct database access, altering an audit entry breaks the cryptographic chain.'**

Function 2.4: Channel Severity Threshold Tuner

[OPTIONAL BONUS]

Source Files: `backend/app/services/pipeline_runner.py, frontend/src/app/policy/page.tsx`

Target Persona: Brand Manager

Function Summary: Allow configuring severity levels per channel (e.g., informal slang is INFO on social media, but CRITICAL in formal press releases).

How It Works: In policy configuration, toggle matrix for channel vs rule severity. When checking copy, the pipeline runner applies channel-specific severity overrides.

Enterprise Benefit: Prevents unnecessary false positives in casual channels while keeping strict guards on investor and press releases.

Judge Pitch Impact: **Shows nuanced enterprise understanding of marketing distribution channels.**

Function 2.5: Exportable Regulatory Audit Dossier (CSV/JSON Bundle)

[OPTIONAL BONUS]

Source Files: `frontend/src/app/transparency/page.tsx, backend/app/api/v1/governance.py`

Target Persona: Legal & Compliance Teams

Function Summary: Add a 1-click 'Export Audit Dossier' button on /transparency that downloads all decisions, policy versions, and override justifications in a standard CSV/JSON package.

How It Works: Compliance officer clicks export to get an audit-ready archive containing timestamps, reason codes, human reviewer identities, and digital signatures.

Enterprise Benefit: Simplifies SEC/FTC regulatory discovery and internal brand compliance audits.

Judge Pitch Impact: **Highlights turnkey enterprise compliance readiness.**

TEAMMATE 3 — Product, Presentation & Demo Experience Lead

Domain Ownership: Judge demo choreography, automated walkthroughs, executive ROI analytics, and enterprise notification integrations.

Function 3.1: 'Judge Pitch Mode' 3-Minute Automated Tour Runner

[CORE REQUIRED]

Target Persona: Hackathon Pitcher & Judges

Source Files: `frontend/src/app/demo/page.tsx`

Function Summary: Enhance /demo with a guided 'Start 3-Minute Judge Pitch' auto-runner that steps through the 4-act story with automated highlights, cue-cards, and synchronized route transitions.

How It Works: Presenter clicks 'Start Live Pitch'. Tour navigates: Act 1: Author catches injection -> Act 2: Reviewer overrides with Double Receipts -> Act 3: Brand Manager backtests and adopts v1.1 -> Act 4: Release Certificate download. Has pause/next/back buttons.

Enterprise Benefit: Guarantees that the presentation will never stutter, skip key features, or get derailed even under high-pressure 3-minute hackathon time limits.

Judge Pitch Impact: **Eliminates demo risk and ensures every single judge sees all 4 personas and all core invariants.**

Function 3.2: Simulated Enterprise Webhook & Slack Notification Dispatcher

[CORE REQUIRED]

Target Persona: Marketing Team & Operations Lead

Source Files: `frontend/src/app/review/page.tsx, frontend/src/app/submissions/page.tsx`

Function Summary: When key governance events occur (Critical Violation detected, Send-Back requested, or Final Certificate issued), display an animated Slack-style banner with the payload preview.

How It Works: Upon granting sign-off, a toast slides in: '■ Sent to #brand-governance Slack: Acme Q1 Campaign officially approved with Cert #CERT-ACME...'. Clicking toast opens the webhook JSON payload.

Enterprise Benefit: Demonstrates how Canon effortlessly fits into existing marketing technology stacks (Slack, Teams, CMS webhook triggers, HubSpot).

Judge Pitch Impact: **Proves enterprise integration feasibility beyond an isolated web application.**

Function 3.3: Executive Governance ROI Scorecard & Cost Calculator

[CORE REQUIRED]

Target Persona: Chief Marketing Officer (CMO) & VP Legal

Source Files: `frontend/src/app/governance/page.tsx`

Function Summary: On /governance, render an Executive Scorecard showcasing First-Time Pass Rate (78.4%), Review Turnaround (1.2 hrs vs 48 hrs baseline), and Estimated Statutory Fines Prevented ($185,000).

How It Works: Interactive sliders allow adjusting monthly campaign volume (e.g. 50 assets/mo) and hourly legal review costs ($350/hr) to calculate dynamic annual ROI savings ($412,000/yr).

Enterprise Benefit: Translates technical rules and linting into indisputable C-suite business justification.

Judge Pitch Impact: **Winning hackathon pitch slide: 'Canon pays for itself in 3 weeks by automating 90% of routine brand compliance reviews.'**

Function 3.4: Quick Persona Switcher Floating Widget

[OPTIONAL BONUS]

Target Persona: Demo Presenter & Tester

Source Files: `frontend/src/components/PersonaSwitcherPill.tsx, frontend/src/app/layout.tsx`

Function Summary: A sticky bottom-right floating pill that displays current user (e.g. Alice Brand Manager) and allows 1-click switching to Charlie Creator or Rachel Reviewer with live permission badge preview.

How It Works: Presenter clicks avatar pill, selects 'Rachel Reviewer (Brand Reviewer)'. Active cookie and role update immediately without visiting /login.

Enterprise Benefit: Makes persona demonstration effortless during judge Q&A; when asked about role separation.

Judge Pitch Impact: **Demonstrates Canon's Zero Automated AI Veto and strict Role-Based Access Control (RBAC).**

Function 3.5: One-Click Demo Environment Reset

[OPTIONAL BONUS]

Target Persona: Demo Presenter

Source Files: `frontend/src/app/Navbar.tsx, backend/app/api/v1/governance.py`

Function Summary: A discreet 'Reset Demo Data' button in the footer or settings that resets test submissions, approvals, and proposals to pristine factory demo state in 1 click.

How It Works: If the team tests or modifies data during practice runs, clicking 'Reset Demo' calls the backend seed endpoint and restores all original fixtures in 500ms.

Enterprise Benefit: Zero panic right before judges walk up; guarantees 100% predictable presentation states.

Judge Pitch Impact: **Safeguards live presentation reliability.**

Master Execution Matrix Summary

Teammate	Core 1	Core 2	Core 3	Optional 1 & 2
Teammate 1 (Frontend)	Printable Certificate PDF	Word/Char Counter & Limits	Cmd+K Command Palette	Revision Diff Visualizer & Audio Scan Cues
Teammate 2 (Policy & Legal)	Claims Expiry Simulator	Statutory Legal Badges	Audit Hash Chain Verifier	Channel Severity Tuner & Audit Dossier Export
Teammate 3 (Product & Demo)	Judge Pitch Auto-Runner	Slack/Webhook Toast	Executive ROI Scorecard	Persona Switcher Pill & Demo Reset Button